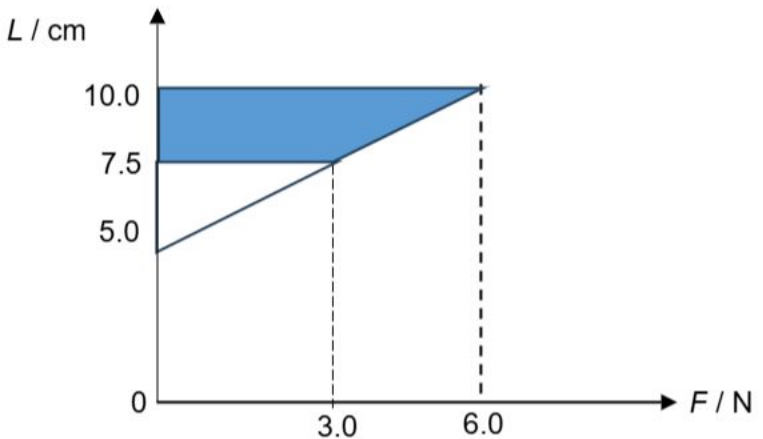
6 The graph below shows how the length L of a spring suspended from one end varies with the force F exerted on the other end.



Initially the spring has an extension of 2.5 cm. What is the work done to extend the spring by a further 2.5 cm?

A 0.11 J **B** 0.15 J **C** 0.26 J **D** 0.45 J

Ans: A

The area representing the work done shoud be with the extension (and in this case the length) axis.

Work done = $\frac{1}{2}(3.0 + 6.0) \times (10.0 - 7.5) \times 10^{-2}$
= 0.11 J